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Abstract

Campus Navigator requires a management view of indoor navigation activity beyond the mobile user's own route. This project develops the backend integration and web-based administration client that expose navigation summaries and named user positions on a shared two-dimensional campus map. The system separates current presence from historical analytics through three Go services: a Presence Gateway, a Trajectory Worker and an Analytics API. Redis maintains active sessions and buffers events, while ClickHouse stores trajectory and journey records for aggregation. The administration client provides two pages. A dashboard summarises navigation starts, journey outcomes and popular destinations for the current day or week using readable place names. A live map displays active users with names and supports limits of 10, 20, 30 or all available markers. An iterative implementation process addressed map-origin alignment, end-to-end display-name propagation and the distinction between representative mobile snapshots and complete administrative snapshots. Local Docker verification used 100 simulated sessions sending positions through the backend rather than generating markers in the browser. The verification demonstrated named moving users and reconciled dashboard counts under that synthetic workload. Deployment scripts and documentation prepare the services and frontend for a Google Compute Engine virtual machine with public HTTPS and private backend HTTP. Public cloud deployment, real-world positioning accuracy and production-scale capacity remain unverified. The resulting prototype provides a focused operational view while retaining clear boundaries around privacy, data durability and future access control.